# AI Agents, AI Harnesses, and AI Tools

Detailed Knowledge Base — Claude • Gemini • Hermes • DeepSeek • Codex

**Purpose:** This RAG document focuses on AI agents, AI harnesses, AI tools, the relationship between agents and harnesses, and five major AI systems/tools: Claude, Gemini, Hermes, DeepSeek, and Codex.

**Important architecture distinction:** An AI agent is primarily concerned with deciding what actions to take toward a goal. An AI harness is primarily concerned with providing and controlling the runtime in which those actions are executed. In real products, these layers may be tightly integrated.

# 1. AI Agents

## Definition

An AI agent is a software system that can interpret a goal, reason about possible actions, use tools, observe results, and continue through multiple steps until it reaches an appropriate stopping condition.

## Agent loop

A common loop is: receive a goal → understand context → decide an action → call a tool or generate a response → observe the result → decide whether another action is needed → finish.

## Reasoning and planning

Agents can break a task into subtasks and select actions. Some agents explicitly create plans; others choose the next action dynamically.

## Tool use

Tools extend an agent beyond language generation. Examples include web search, databases, APIs, file access, code execution, and retrieval systems.

## State and memory

Agents may maintain conversation history, task state, retrieved information, intermediate results, and persistent information depending on the design.

## Autonomy

Autonomy describes how much action selection is delegated to the system. A simple assistant may perform one action, while a more autonomous agent may perform many steps with limited human intervention.

## Stopping conditions

An agent should stop when the task is complete, a final answer is ready, a step limit is reached, a timeout occurs, or required information is unavailable.

## Limitations

Agents can misunderstand requests, choose inappropriate tools, hallucinate, provide invalid arguments, or enter loops. Execution controls and evaluation help identify these problems.

# 2. AI Harnesses

## Definition

An AI harness is the runtime, orchestration, and control infrastructure surrounding an AI model or agent. It manages how the system executes tasks rather than being the model itself.

## Purpose

A harness provides a controlled environment where an agent can access tools and resources. It can manage state, context, permissions, errors, logging, and evaluation.

## Common responsibilities

Responsibilities may include session management, context handling, tool registration, tool execution, authentication, authorization, retries, timeouts, sandboxing, logging, tracing, output validation, cost tracking, and benchmarking.

## Harness and agent relationship

The agent decides what should happen next toward a goal. The harness provides the environment, tools, constraints, and execution mechanisms needed to carry out those decisions.

## Harness without a fully autonomous agent

A harness can orchestrate deterministic workflows or direct LLM calls, so a harness does not necessarily require a fully autonomous agent.

### Guardrails

Guardrails restrict or validate behavior through permissions, input validation, output schemas, maximum steps, timeouts, authentication, and approval requirements.

### Observability

A harness can record model calls, tool calls, tool results, timing, errors, and final outputs. This supports debugging and benchmarking.

### Evaluation

A harness can run standardized tasks repeatedly and collect accuracy, success rate, latency, tokens, tool-call count, failures, and cost.

## 3. AI Agent vs AI Harness — Detailed Difference

### Core purpose

An agent is designed to pursue a goal by deciding actions. A harness is designed to operate and control the AI system that performs those actions.

### Decision-making

The agent is normally responsible for deciding which action or tool may be useful. The harness normally enforces whether and how that action can actually execute.

### Execution

The agent may request an action; the harness or runtime executes the requested operation through the available tool or environment.

### Tools

The agent chooses among available tools based on the task. The harness registers, exposes, authenticates, authorizes, executes, and monitors those tools.

### State

An agent can use task state to reason about what has happened. The harness can store and manage the state needed for reliable execution.

### Permissions

An agent may request a sensitive operation, but the harness can reject it when permissions or policies do not allow the operation.

### Errors

An agent may recognize that a tool result is incorrect or incomplete. The harness can provide retries, timeouts, error handling, and execution recovery.

### Observability

The agent produces actions and results as part of its work. The harness can record traces of those actions and results for debugging and evaluation.

### Evaluation

The agent is the behavior being evaluated. The harness can repeatedly run tests and collect metrics about that behavior.

### Relationship

The agent and harness are complementary. The agent supplies goal-oriented decision-making; the harness supplies controlled execution and orchestration.

### Can they exist separately?

Yes. An LLM or agent can be used with a simple custom runtime, and a harness can orchestrate a workflow without requiring a highly autonomous agent.

### Simple analogy

The agent can be thought of as the decision-maker, while the harness is the controlled workspace and operating environment that gives the decision-maker capabilities and limits.

## 4. Agent vs Harness — Example

### Scenario

Suppose a user asks: “Find information about AI harnesses and summarize it.”

### Agent behavior

The agent interprets the goal, decides that search or retrieval is needed, selects an available search/retrieval tool, examines the returned information, and decides what to include in the final response.

### Harness behavior

The harness provides the search/retrieval tool, authenticates access, executes the tool call, applies permissions and limits, records the call, handles errors, and returns the result to the agent.

### Result

The agent determines what should be done; the harness controls how the requested action is executed. They work together but have different responsibilities.

## 5. AI Tools

### Definition

An AI tool is a callable capability that performs a specific operation for an AI system.

### Tool categories

Examples include web search, database queries, APIs, code execution, file operations, document retrieval, calculators, communication services, and business-system integrations.

### Tool schema

A tool normally has a name, description, input parameters, and output. Structured schemas help the model produce valid arguments.

### Tool calling

The model or agent can return a structured request containing a tool name and arguments. The runtime executes the tool and returns the result.

### Tool selection

The agent chooses among available tools based on the task, descriptions, current state, and permissions.

### Tool safety

Sensitive operations should use authentication, authorization, validation, and appropriate human approval.

## 6. Claude

### What is Claude?

Claude is Anthropic's family of AI models and assistant products. Claude can be used conversationally and through developer APIs.

### Claude as an agent component

Claude can serve as the reasoning and generation model inside an agent architecture. An external harness can provide tools, context, state, permissions, and execution.

### Tool use

Anthropic provides tool-use capabilities through its developer platform. Tools can be described with schemas, allowing Claude to request external operations.

### Advanced tool use

Anthropic has documented advanced mechanisms such as Tool Search Tool, Programmatic Tool Calling, and Tool Use Examples for tool-rich workflows.

### Agent distinction

Claude is primarily a model/platform ecosystem rather than a single definition of an autonomous agent. A complete Claude-based agent may consist of Claude plus a harness, tools, retrieval, memory, and an execution environment.

### Coding workflows

Claude can be used in coding-agent environments where the surrounding system provides files, terminals, repositories, and other capabilities.

### Considerations

Capabilities depend on the model version, API, tools, prompts, and runtime. Tool access does not mean unrestricted system access.

## 7. Gemini

### What is Gemini?

Gemini is Google's family of AI models and development ecosystem. Gemini models support text and, depending on the model, additional modalities.

### Gemini as an agent model

Gemini can serve as the reasoning and generation component of an agent. A surrounding framework can provide tools, state, retrieval, and execution.

### Managed agents

Google documents managed agents that can operate in a Linux sandbox, execute code, manage files, and browse the web.

### Agent frameworks

Google documents agent development using frameworks and SDKs such as Google ADK and integrations with other agent frameworks.

### Tool use

Gemini-based agents can be connected to tools and external data sources to perform operations beyond language generation.

### Harness perspective

A Gemini-based agent can use a separate harness or a managed environment that handles orchestration, sandboxing, tools, state, and execution.

### Important distinction

Gemini is primarily a model and platform ecosystem. A particular Gemini agent product can combine model, tools, runtime, and harness.

# 8. Hermes Agent

### What is Hermes?

Hermes Agent is an autonomous AI agent project from Nous Research. It is designed around agent behavior rather than being only a language-model API.

### Agent orientation

Hermes includes agent loops, tools, skills, persistent knowledge, conversation search, and other runtime behavior.

### Learning loop

Hermes documentation describes a learning loop where the agent can create skills from experience and improve behavior during use.

### Skills

Skills are reusable capabilities or instructions that help Hermes perform particular task types.

### Tools

Hermes can use tools and terminal-oriented capabilities depending on its configuration and environment.

### Model flexibility

Hermes can work with different model providers and endpoints, separating the agent layer from the underlying model layer.

### Harness perspective

Hermes contains significant runtime functionality around models, including tools, skills, memory, and agent execution, making it a useful example of an integrated agent/harness architecture.

# 9. DeepSeek

### What is DeepSeek?

DeepSeek provides AI models and APIs. Its developer platform supports structured tool calls and agent-oriented integrations.

### Tool calling

DeepSeek APIs support function tools. The model can return a structured function call while the application or runtime executes the actual function.

### Thinking and tools

DeepSeek documentation describes tool calls during thinking-mode interactions, allowing multiple reasoning and tool-call turns.

### Model layer

DeepSeek can serve as the model behind an agent or coding assistant. A separate harness can provide tools, memory, state, permissions, and execution.

### DeepSeek Harness

DeepSeek documentation describes DeepSeek Harness as a developer-preview agent harness for harness developers. This distinguishes the model/API layer from the harness layer.

### API compatibility

DeepSeek documents OpenAI-format and Anthropic-format API compatibility for supported use cases.

### Tool choice

Supported API modes can provide tool-choice behavior such as none, auto, and required.

### Important limitation

A generated tool call does not itself execute a function. The surrounding application or harness must implement and execute the requested function.

## 10. Codex

### What is Codex?

Codex is OpenAI's AI coding-agent system designed to perform software-engineering tasks such as feature work, refactoring, migrations, and other coding tasks.

### Codex as an agent

Codex operates in an agentic environment where coding tasks can involve files, commands, tools, tests, and longer-running execution.

### Codex harness

OpenAI's Agents API documentation describes access to a Codex harness. OpenAI manages sessions, orchestration, context compaction, and recovery while applications provide tools and choose execution environments.

### Sandbox execution

Codex agents can operate in sandbox environments where they can execute code, edit files, connect to MCP servers, and produce artifacts, depending on configuration.

### Multi-agent workflows

Codex supports multi-agent workflows in supported environments, allowing different agents to work on separate tasks or parts of a project.

### Skills

Codex Skills can provide team-specific standards, workflows, and procedures for repeated engineering tasks.

### Background work

Codex can support longer-running software tasks such as issue triage and other engineering workflows.

### Harness perspective

Codex is an example of an integrated coding-agent architecture where model capabilities, tools, sandboxing, orchestration, context management, and recovery work together.

## 11. Comparison of the Five Systems

### Claude

Primarily a model/platform ecosystem. It can operate as the reasoning model inside external agent harnesses and supports structured tool use.

### Gemini

Primarily a model/platform ecosystem with agent frameworks and managed agent environments.

### Hermes

An explicit autonomous agent project with tools, skills, memory, learning behavior, and model-provider flexibility.

### DeepSeek

Primarily a model/API ecosystem with function calling and tool-enabled reasoning, plus a documented DeepSeek Harness developer preview.

### Codex

An AI coding-agent and harness ecosystem focused on software engineering, with tools, sandboxing, orchestration, and context management.

## 12. Benchmarking Agents and Harnesses

### Accuracy

Measures whether the answer or task result is correct according to predefined criteria.

### Success rate

Measures the percentage of tasks meeting the defined success conditions.

### Execution time

Measures total task completion time, potentially including model, retrieval, tool, and network time.

### Tokens

Measures input and output token consumption.

### Tool calls

Measures how many external operations were performed.

### Failure rate

Measures tasks that fail, time out, return invalid results, or do not meet criteria.

### Cost

Measures model and service costs for the tested task.

### Consistency

Measures whether repeated runs behave similarly under comparable conditions.

### Fair comparison

Use the same tasks, conditions, tool availability, success criteria, and measurement method when comparing systems. Benchmark results are dependent on the test design.

## 13. Practical Q&A;

### What is an AI agent?

A system that can interpret a goal, select actions, use tools, observe results, and continue until an appropriate stopping condition.

### What is an AI harness?

The runtime and orchestration infrastructure that manages execution around an AI workflow or agent.

### What is the main difference between an agent and a harness?

The agent is focused on deciding what to do toward a goal; the harness is focused on controlling and executing the environment in which those decisions occur.

### What is an AI tool?

A callable external capability that performs a specific operation for an AI system.

## Is Claude an agent?

Claude is primarily a model and product family. It can be used as the reasoning component of an agent and can participate in tool-using agent systems.

## Is Gemini an agent?

Gemini is primarily a model family and ecosystem. Google also provides managed agent capabilities around Gemini.

## Is Hermes an agent?

Yes. Hermes Agent is explicitly an autonomous agent project.

## Is DeepSeek an agent?

DeepSeek primarily provides models and APIs, while its ecosystem supports agent tool use and includes a DeepSeek Harness developer preview.

## Is Codex an agent?

Codex is an AI coding-agent system designed to perform software-engineering tasks in an execution environment.

## Can an LLM work without an agent?

Yes. A model can directly answer a question without autonomous planning or tool selection.

## Can a harness work without an autonomous agent?

Yes. A harness can orchestrate deterministic workflows or direct model calls.

## Can an agent use multiple tools?

Yes. An agent can be provided with multiple tools and choose among them based on the task and available permissions.

# 14. Compact Agent vs Harness Reference

| Aspect | AI Agent | AI Harness |
|---|---|---|
| Primary role | Decides actions toward a goal | Controls and orchestrates execution |
| Decision-making | Selects next action/tool | Enforces what can execute |
| Tools | Chooses when tools are useful | Registers, executes, and controls tools |
| State | Uses state to reason about the task | Manages execution/session state |
| Errors | Interprets failed results | Retries, times out, logs, or stops execution |
| Permissions | Requests an action | Enforces authorization and limits |
| Evaluation | Behavior being evaluated | Runs tests and collects metrics |

# 15. Recommended Dify Grounding Rule

For questions about AI agents, AI harnesses, AI tools, Claude, Gemini, Hermes, DeepSeek, or Codex, use the retrieved knowledge base context as the primary source. Do not invent information. Clearly distinguish a model from an agent, a harness, and a tool. If the retrieved context does not contain enough information, say that the knowledge base does not contain enough information to answer the question.

# 16. Suggested RAG Test Questions

- What is the difference between an AI agent and an AI harness?
- What responsibilities belong to an AI harness?
- What responsibilities belong to an AI agent?
- How does an agent use tools?
- What is Claude and how does it fit into an agent architecture?
- What is Gemini and how does it fit into an agent architecture?
- What is Hermes Agent?
- What is DeepSeek Harness?
- What is Codex and what does its harness do?
- What are the main differences between Claude, Gemini, Hermes, DeepSeek, and Codex?
- Can an LLM work without an agent?
- Can a harness operate without an autonomous agent?
- How can AI agents be benchmarked?